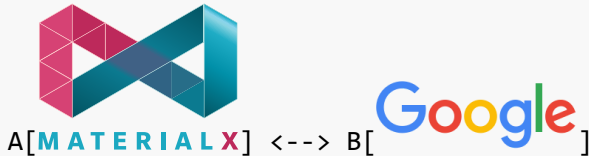


MaterialX Protobuf Converter

A sample Python utility for converting MaterialX documents to and from Protocol Buffer format.

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What is Protocol Buffers?

[Protocol Buffers \(protobuf\)](#) is Google's mechanism for serializing structured data.

Benefits of Protocol Buffer Format

- **Compact:** Binary format is smaller than XML
- **Fast:** Efficient serialization and deserialization
- **Cross-platform:** Language-agnostic format works across different platforms and programming languages
- **Typed:** Strong typing for data integrity
- **Versioning:** Built-in support for schema versioning and backward compatibility

Overview

This project provides tools to:

- Convert MaterialX XML documents to Protocol Buffer format
- Serialize MaterialX documents to binary `.mxpb` files
- Export MaterialX documents to JSON format
- Convert Protocol Buffer documents back to MaterialX XML
- Generate visual diagrams of MaterialX document structure

Prerequisites

macOS Setup

Install Protocol Buffers compiler:

```
brew install protobuf
```

Windows Setup

Install Protocol Buffers compiler:

```
choco install protoc
```

Linux Setup

Install Protocol Buffers compiler:

```
sudo apt-get install -y protobuf-compiler
```

Python Setup

1. Install Python protobuf library:

```
pip install --only-binary=all protobuf
```

2. Install MaterialX Python library:

```
pip install MaterialX
```

Protocol Buffer Schema

The schema defines two main message types:

- **Attribute:** String key value pair. This is added to allow usage of repeated attributes as opposed to usage of maps. Maps in ProtoBuf are ordered which is undesirable for MaterialX where attribute order matters for types such as node inputs and outputs as these define the signature of the node.
- **MaterialXElement:** Represents a MaterialX element with name, type, attributes, and child elements.
- **MaterialXDocument:** Represents the root of the element hierarchy and contains top-level attributes.

This schema reflects the simplicity of the MaterialX representation.

A schema **Version** message is also provided for version management including allowing for an upgrade mechanism if ever required.

The schema is defined in the `materialx.proto` file.

Generating Python Code from Proto Schema

Using the `materialx.proto`, Python bindings can be generated with the following command:

```
protoc --python_out=. materialx.proto
```

This generates `materialx_pb2.py` which contains the Protocol Buffer message classes. These classes are used in the conversion logic to serialize and deserialize MaterialX documents.

File Structure

```
- materialx.proto          # Protocol Buffer schema definition
- materialx_pb2.py         # Generated Python protobuf bindings
- materialx_serializer.py  # Conversion logic between MaterialX and
Protobuf
- main.py                 # Command-line utility for conversion
- *.mtlx                  # Sample MaterialX documents
```

Usage

MaterialX to Protobuf Conversion

Convert a MaterialX document to Protobuf format:

```
python main.py path/to/<document>.mtlx
```

This command can optionally generate three output files:

1. `.json` - JSON representation
2. `.mxpb` - Binary Protocol Buffer format (compact serialization)
3. `.mmd` - Mermaid diagram
4. `_converted.mtlx` - Round-trip conversion back to MaterialX XML (for validation)

Example

```
python main.py standard_surface_chess_set.mtlx
```

Outputs:

- `standard_surface_chess_set.json` - JSON format
- `standard_surface_chess_set.mxpb` - Binary protobuf
- `standard_surface_chess_set_converted.mtlx` - Converted back to MaterialX

Protobuf to MaterialX Conversion

Convert a Protobuf binary file back to MaterialX XML:

```
python main.py path/to/<document>.mxpb
```

This will generate a MaterialX XML file named `<document>_from_pb.mtlx`.

This command can optionally generate three output files:

1. `.json` - JSON representation
2. `.mmd` - Mermaid diagram

Core Components

`materialx_serializer.py`

MaterialXToProtobuf

Converts MaterialX documents to Protocol Buffer format:

- Extracts document-level attributes
- Traverses the MaterialX document
 - Extracts element names, types, and attributes
 - Recursively converts child elements

ProtobufToMaterialX

Converts Protocol Buffer format back to MaterialX:

- Creates MaterialX document
- Reconstructs MaterialX document attributes
- Restores element hierarchy and attributes

Util

Utility functions for debugging and visualization:

- `from_string()` - PB doc from string
- `to_string()` - PB doc to string
- `to_json()` - PB doc to json
- `generate_mermaid_diagram()` - Generate Mermaid diagrams for visualization

Sample Files

The repository includes sample MaterialX documents:

- `standard_surface_chess_set.mtlx` - Chess set with standard surface shader
- `unlit_cross.mtlx` - Simple unlit cross shader

Each sample includes its converted outputs as appropriate (`.json`, `.mxpb`, `_converted.mtlx`, `from_pb.mtlx`).

Visualization

Enable Mermaid diagram generation in `main.py` by setting `write_mermaid = True`. This generates a `.md` file with a Mermaid diagram showing the document structure.